

Project Documentation Notice: This document is created specifically for the Week 7 IoT, Connectivity & RTOS portfolio project. It details the cloud NoSQL database schema, real-time synchronization, security rules, and client-side data export implemented in Tasks 4 and 5.

1. Cloud Database Architecture

Firestore Realtime Database is a cloud-hosted NoSQL JSON document store. Data is synchronized in real time to every connected client via persistent WebSockets and HTTP Server-Sent Events (SSE). In this project, the database acts as the single source of truth connecting embedded ESP32 nodes and the frontend web dashboard.

2. Project NoSQL Data Schema

The database root contains three primary nodes:

- `/live_telemetry` : Stores the latest immediate reading (temperature, humidity, ldr, timestamp) overwritten on each cycle.
- `/control` : Stores actuator setpoints (bulbState: boolean, autoMode: boolean, threshold: integer).
- `/logs` : An append-only chronological array where records are queued with unique chronological push IDs.

3. Security Rules & Access Control

The database uses declarative security rules to protect telemetry streams while allowing authenticated dashboard clients to update control registers:

Read access is public for the dashboard visualization, while write permissions require an active authenticated user token or validated device credentials.

4. Client-Side Analytics & CSV Extraction

Historical telemetry records stored under `/logs` are serialized into Comma Separated Values on demand using the browser native Blob and URL.createObjectURL APIs. Records are formatted with standard CRLF line breaks to guarantee full compatibility with Microsoft Excel, Apple Numbers, and Google Sheets.